NAVEEN KUMAR REDDY SOMIREDDY

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Research CV · PhD applications

PROFILE

Early-career machine learning researcher focused on federated learning, continual learning, and machine unlearning. Translates research ideas into working algorithms, builds experimental frameworks and baselines, designs controlled experiments, and automates large-scale evaluations, across both the implementation and the theoretical development of the methods. Co-author of FedQCL (CoLLAs 2026, oral presentation, top 8% of accepted papers) and DUET-CLU (under review, AISTATS 2027). Seeking a PhD to develop reliable learning systems that adapt, retain, and forget, through algorithmic innovation, rigorous empirical analysis, and principled optimization.

EDUCATION

Indian Institute of Engineering Science and Technology (IIEST), Shibpur 2021–2025
B.Tech. in Information Technology | CGPA: 8.68/10
Relevant coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Linear Algebra, Operating Systems, DBMS, Computer Networks, Computer Architecture

PUBLICATIONS

[C1] Federated Continual Learning as a Distributed Drift-Plus-Penalty Control Problem CoLLAs 2026 (Oral)
Nazreen Shah, Naveen Kumar Reddy Somireddy, Zubair Shaban, Ranjitha Prasad, Bharath B. N. Accepted; oral presentation (top 8% of accepted papers). Also accepted at the NeurIPS 2026 WiML Workshop. arXiv:2608.21539

PREPRINTS & MANUSCRIPTS

Unified Continual Learning and Unlearning with Adaptive Constraints (DUET-CLU) Under review · AISTATS 2027
Nazreen Shah, Prashant Khanduri, Naveen Kumar Reddy Somireddy, Bharath B. N., Ranjitha Prasad.

FedResilience: Concurrent Handling of Client Dropout and Slow Learners in Federated Learning Manuscript
Naveen Kumar Reddy Somireddy (first author), Sachin Dudda Nagaraju, Annappa B. Proximity-based weight transfer and adaptive batch scheduling.

RESEARCH EXPERIENCE

Research Assistant, Federated Continual Learning & Unlearning Dec 2024–Mar 2026
Indraprastha Institute of Information Technology Delhi (IIIT Delhi) | Advisor: Prof. Ranjitha Prasad

FedQCL: Federated Continual Learning via Drift-Plus-Penalty Co-author · CoLLAs 2026 (Oral)

- Studied catastrophic forgetting and client drift by formulating federated continual learning as a stochastic control problem. FedQCL uses client-local virtual queues to track accumulated forgetting and a tunable parameter to balance stability and plasticity, without gradient projection or extra communication.
- With ResNet-18 and five non-IID clients, reported 54.5% vs. 51.0% for DER++-FL on Split-CIFAR-100, 85.7% vs. 83.9% on Split-CIFAR-10, and 33.4% vs. 31.0% on Split-TinyImageNet (Task-IL). Class-IL on Split-CIFAR-100: 24.92% vs. 24.32% (near-IID) and 14.66% vs. 14.02% (non-IID) against DER++.

My contributions

- Implemented the FedQCL method and its full experimental codebase; implemented or integrated all baselines (FedAvg, FedProto, EWC-FL, LwF-FL, ER-FL, DER-FL, DER++-FL, CCVR, FOT, BI); contributed to the theoretical development of the framework.
- Designed experiments and ablations, tuned hyperparameters, and ran the evaluation suite across data heterogeneity, buffer size, client count, and task count in Task-IL and Class-IL settings; produced the results reported in the paper.
- Code: github.com/Naveensomireddy4/FedQCL_Task · github.com/Naveensomireddy4/FedQCL_CIL

DUET-CLU: Unified Continual Learning and Unlearning Co-author · AISTATS 2027 (under review)

- Formulates retention and non-resurgence as task-level constraints in a primal-dual framework with adaptive slack variables and dual multipliers; includes a dual-sensitivity characterization and a slack-dual equilibrium under convex per-request assumptions.

My contributions

- Implemented the DUET-CLU framework and its experimental codebase; implemented or integrated the comparison baselines (Retrain, EWC, LwF, ER, DER++, LSF, CLPU, CLMUL, UnCLe, PALL); contributed to the theoretical formulation and analysis.
- Designed experiments and ablations, tuned hyperparameters, and evaluated the effects of slack variables and dual multipliers on continual learning and unlearning in Task-IL and Class-IL settings.

Continual Learning for Federated Speech Models (Whisper-small)

- Built a federated training pipeline for Whisper-small and integrated A-GEM as the continual-learning method; reported 96% accuracy (IID), nearly matching centralized training on all the data, and 80% (non-IID).

Research Intern, Federated Learning May 2024–Nov 2024

National Institute of Technology Karnataka (NITK), Surathkal | Advisor: Prof. Annappa B.

FedResilience: Client Dropout and Slow-Learner Resilience First author

- Proposed a FedAvg-based framework combining proximity-based weight transfer from the most similar active client (cosine similarity) for dropout handling with adaptive dual-batch scheduling for slow learners.
- On BloodMNIST with ResNet-18 under non-IID data, reached 82.1% accuracy in 8 rounds, versus 80.95% for FedAvg in 15 rounds.

INDUSTRY EXPERIENCE

Data Analyst & Data Engineer (MLOps), Tredence, Bengaluru Apr 2026–Oct 2026

- Built automation for creating and running data pipelines, and used it to migrate five pipelines (20 tables; ~3 billion records each) from Hive to GCP: converted SQL jobs to Spark (Scala), automated Dataproc runs, and moved reporting to Power BI.
- Automated pipeline setup and execution, reducing manual effort and making runs repeatable.

SELECTED PROJECT

Real-Time Scene Graph Generation Jan 2024–Mar 2024

YOLOv5, PAN-FPN, GloVe, PyTorch | Advisor: Dr. Arindam Biswas

- Built a real-time scene-graph model for object detection and relation classification using PAN-FPN feature fusion, spatial encodings, and ROI Align; 14.5 mR@100 at 30 FPS on the Panoptic Scene Graph dataset.

TECHNICAL SKILLS

Research: Federated learning, continual learning, machine unlearning, constrained optimization, primal-dual methods, experiment design and automation
Programming & ML: Python, C++, SQL, PyTorch, TensorFlow, Hugging Face
Data & infrastructure: Hive, Spark, Scala, GCP, Power BI, MLOps, Docker, Git, Linux
Models & benchmarks: ResNet-18, Whisper; Split-CIFAR-10/100, Split-TinyImageNet, BloodMNIST
Baseline library: FedAvg, FedProto, EWC, LwF, ER, DER/DER++, A-GEM, CCVR, FOT, BI, LSF, CLPU, CLMUL, UnCLe, PALL (github.com/Naveensomireddy4/Baselines)

HONORS

- CoLLAs 2026: Oral presentation; top 8% of accepted papers.
- LeetCode: Knight (top 3%); 1,100+ problems solved; Rank 97 in Biweekly Contest 133.

REFERENCES

Prof. Ranjitha Prasad (IIIT Delhi) · Prof. Annappa B. (NITK Surathkal). Contact details available on request.